Structural Mechanics: Directional Structural Witnesses for Fail-Closed Cross-Domain Transfer

Sze Chun Yiu
Stockholm University
sze-chun.yiu@fysik.su.se

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Abstract

Orion can persist task episodes, consolidate scoped lessons and reuse experience without changing the underlying model weights, but the existence of a shared experience substrate does not determine when cross-domain reuse is valid. We study the transfer relation itself. A directional structural witness records context, quantity of interest, role mappings, preserved invariants, non-preserved properties, target boundaries, evidence and uncertainty, and is intended to license or reject movement of an episode, lesson or operator into a new problem fibre. In the deployed obstruction–transformation router, this witness supplies the transport obligation for cross-domain `JUMP` and for transported components used by `GLUE`: retrieval may nominate a source, but complete source-precondition accounting, disanalogies, forbidden-loss checks and target validation determine whether it is actually reusable. The executable v2 contract makes these obligations non-compensatory and separates `LICENSED`, `REJECTED` and `CANNOT_CHECK` rather than turning missing applicability evidence into a negative or positive scalar score.

We first preserve the historical constructed diagnostics: a six-case deterministic Q2/Q3 conformance suite and a 44-pair, 11-family leave-one-family-out study in which witness features improved ROC-AUC by 0.086 and average precision by 0.097 over frozen lexical/tag controls. Because those labels were internally proposed, we then registered a separate exact-verifier lane whose gold decision is obtained by executing target-world rules rather than by annotator judgement or perturbation identity. After a first development generator failed the preregistered circularity attack and was discarded, the redesigned development instrument fixed a 0.05 paired-Brier MDE and froze a fresh 576-case confirmatory seed before outcome access. The confirmatory packet comprised 256 licensed, 256 rejected and 64 `CANNOT_CHECK` cases across flow, logic, dimensional-transform and finite-state families. A derived-effect/mechanism-only control reached exact three-way accuracy 0.875 but falsely licensed 25% of verifier-invalid transfers; a relation/invariant-only control reached 0.819 with 37.5% false acceptance. The full applicability contract matched all 576 verifier decisions, retaining every valid transfer, licensing no invalid transfer and abstaining on every verifier-unknown case. Relative to the mechanism-only control, paired binary Brier improved by 0.120 with item-bootstrap 95% interval $[0.0938, 0.1481]$. The residual was positive in all four primary families, but four family clusters are insufficient for a broad cross-domain law, so a separately frozen six-family robustness extension remains required.

For the natural-domain lane we froze a fresh 16-item v2.1 source set, opaque annotation packet and exact content-bound `BAAI/bge-reranker-v2-m3` semantic control before external labels. Label-blind descriptor jobs 3476527–3476529 reached `HARVEST_DESCRIPTOR_READY`. The later operator override closed #217 through an explicitly demoted `AI_OPERATOR/NON_INDEPENDENT` route; after failed environment runs 3476750/3476751 were preserved as negative history, hardened demoted train/inference jobs 3476753/3476754 passed. This is operational demoted evidence only: it does not satisfy independent-human annotation, adjudication, provenance-audit or confirmatory authority. The independent-human residual was frozen as unavailable under #332 and continues under the versioned successor lineage. Moreover, the pre-label power design classifies the frozen $n = 16$ packet as power-limited for the registered

0.05 paired Brier-reduction MDE, for which the simulation indicated roughly $n \approx 48$ items across the modeled noise envelope. The objective known-world result therefore supports the applicability representation, not a claim that a model can infer it from raw science. Strong external semantic/LLM comparators, a disjoint downstream-routing study and independent natural-domain validation remain open before any universal transfer, end-to-end research-utility or learned-witness claim.

1 Problem: repeated payment for reusable structure

Language-model learning and inference both contain repeated work. During training, semantically different examples can instantiate the same dependency, conservation law, feedback motif, constraint pattern or reasoning routine. During inference, an agent can repeatedly reconstruct a procedure that was already solved in earlier trajectories. The tempting conclusion is that a system should externalize the repeated structure and reuse it. That conclusion is not novel. The research problem is to determine *what counts as the same reusable structure, when transfer is valid, and when the cost of inducing and verifying the structure is actually lower than relearning or rediscovering it.*

Several parent methods already occupy large parts of this space. Skill-It learns skill dependencies and uses them to adapt data sampling; in continual pretraining it reports a 3B-parameter model reaching higher evaluation-harness accuracy with 1B tokens than a uniform data-source baseline trained with 3B tokens [1]. MASS constructs a mathematical skill graph for pretraining-data selection and reports comparable downstream mathematical performance after removing substantial fractions of the training tokens [2]. These results make "graph-guided data selection saves training data" an inherited mechanism, not a Orion contribution.

The inference side is similarly mature. Reasoning Primitive Induction mines successful ReAct traces, clusters recurrent moves and compiles them into typed pseudo-tools that improve several held-out reasoning and planning tasks at lower average inference cost [10]. TraceCompiler mines noisy tool-use traces into mostly deterministic workflows with evidence-bearing producer–consumer dependencies and can refuse compilation when an irreversible branch is underdetermined; importantly, it reports call reduction without claiming net efficiency because offline compilation cost was not measured [11]. SWIFT directly frames agent-workflow design as amortized transfer from structural priors and reports large reductions in marginal per-task workflow-search cost across source and unseen tasks [9]. Its operator-name randomization result further suggests that topological structure, not just semantic labels, carries much of the transfer signal.

Abstraction itself is also an active target. Controlled ACL experiments study whether language models learn structural information and can compose it at test time [12]; RLAD trains an abstraction generator and abstraction-conditioned solver rather than relying only on long brute-force traces [20]; related abstraction-reinforcement work reports improved robustness and out-of-distribution reasoning [21]. The candidate contribution here is therefore not "LLMs need abstraction."

1.1 Residual research question

The residual is a conjunction not established by the parents above. We ask whether one persistent structural object can simultaneously support:

1. cross-domain training-data redundancy judgments when surface semantics differ;
2. test-time retrieval or adaptation of a previously learned operator;
3. explicit rejection of semantically similar but structurally invalid analogies;

4. boundary- and QoI-scoped transfer rather than global similarity;
5. evidence-bearing failure history so negative transfer changes future reuse; and
6. total cost accounting that includes the one-time induction and verification cost.

This makes the paper mechanism-seeking rather than leaderboard-seeking. If a skill graph, workflow prior or semantic retriever already supplies the same predictive transfer signal at lower cost, the Orion structural layer has not earned its complexity. Conversely, a positive result is strongest when the same structural identifier is learned or stored once and measurably improves both data selection and later inference on domains whose surface vocabulary would not have linked them directly.

1.2 Terminology at a glance

For readers from machine learning or systems engineering, the following plain-language glosses fix the vocabulary used throughout; formal definitions appear at first technical use.

Quantity of interest (QoI)
: the specific target quantity a transfer is meant to preserve; correctness is judged relative to it, not to overall similarity.

Directional structural witness
: the typed object that licenses moving structure from a source to a target: a role map, the invariants that must be preserved, the properties that are *not* preserved, target boundary conditions, and evidence.

Obligation set
: the witness expanded into an executable checklist of transfer obligations, each evaluated `SATISFIED` / `VIOLATED` / `UNKNOWN`.

Non-compensatory
: a single violated load-bearing obligation forces rejection; strength elsewhere cannot buy it back — there is no weighted average.

`LICENSED` / `REJECTED` / `CANNOT_CHECK`
: the three fail-closed verdicts: transfer allowed, transfer refused on a violation, or abstention when required evidence is missing (`UNKNOWN` $\neq$ `VIOLATED`).

Fibre
: the retrieval index/representation scoped to a fixed (QoI, context) into which an episode is moved.

Forbidden loss
: a property whose loss under transfer invalidates the reuse even if the mapping otherwise fits.

Demoted evidence
: an evidence path deliberately denied confirmatory/independent authority (e.g. an AI operator standing in for an absent independent annotator).

Structural-OOD
: out-of-distribution by structure: a known structure in a new domain, or a whole structure family held out.

MDE
: minimum detectable effect — the smallest effect size a frozen test is powered to detect.

TPVS; break-even count
billable inference tokens per *valid* solve (a solve that also passes the transfer/verification gates); and the smallest reuse count at which the structural layer is no more costly than the comparator.

1.3 Shared experience banks and dependency-aware skill retrieval narrow the residual further

A closer parent for the "learn once, reuse across tasks" intuition is ReX, which treats recurring reasoning patterns and skills as latent experiences rather than task-specific adapters [13]. ReX learns a shared Experience Bank of foundational skill vectors, encodes each input into a low-dimensional experience code, and dynamically composes the relevant skill vectors into a lightweight adapter without requiring explicit task identifiers. Its multi-task results show that cross-input/task experience reuse can improve specialization and generalization. This removes any novelty claim that Paper 3 uniquely proposes a persistent shared experience substrate or dynamic composition of reusable skills.

Dependency-aware skill retrieval is also moving beyond flat semantic matching. SkillGraph represents agent skills as a dependency graph and targets selection/composition failures that arise when modular skills are treated as isolated text descriptions [14]. SKILLGRAPH goes further by representing reusable skills as an evolving typed graph whose prerequisite, enhancement and co-occurrence edges are updated from trajectories and reinforcement-learning feedback; ordered skill subgraphs guide new-task execution while the same graph participates in improving the policy [15]. GraSP compiles retrieved skills into executable typed DAGs with precondition–effect edges, node-level verification and locality-bounded repair [16]. AgentGL similarly treats graph topology as an active reasoning substrate: an LLM agent uses graph-native tools at inference while a graph-conditioned curriculum is used during reinforcement learning [18]. These systems are important parent mechanisms because they show that relational topology can inform learning, retrieval, composition and execution.

A complementary retrieval parent exposes why the negative controls must be strong. SkillSight shows that recurring descriptive boilerplate across skill documents can inflate dense and lexical relevance scores even when those shared words are not discriminative of the required capability; calibrating that background improves retrieval without extra model training [17]. Paper 3 therefore cannot treat a semantic retriever failure as evidence for structural novelty unless the semantic baseline is already calibrated against shared-background bias.

The candidate residual must therefore remain more specific. Paper 3 asks whether a *scientifically scoped* structural object—with explicit QoI, context, directional role mapping, preserved and non-preserved properties, boundary conditions, evidence and negative-transfer history—can do two jobs with the same persistent identity: estimate *cross-domain training-data redundancy* and license or reject inference-time transfer. ReX supplies reusable latent experience; SkillGraph/SKILLGRAPH/GraSP supply increasingly rich dependency and execution graphs; AgentGL demonstrates one graph-conditioned learning/execution loop; SkillSight supplies a stronger semantic-retrieval control. Orion must either add value in the controlled surface-disjoint, boundary-sensitive training-selection plus inference regime or narrow its claim to a scientific-validity governance layer rather than a general reuse advance.

Post-v3 residual. Orion now implements the persistent episode/lesson/tool substrate that earlier drafts treated as part of the broader motivation. That implementation narrows the paper rather than strengthening its empirical claim. The remaining question is whether the directional witness is a useful *transfer-validity relation* inside such a substrate. Persistence, lesson consolidation and shared retrieval are therefore architectural premises supplied by the framework; boundary-sensitive cross-domain licensing remains the scientific object tested here.

2 A scoped structural transfer object

A reusable structure should not be a bag of similar words. Let a structural object be

$$S = (V, E, \chi, \gamma, q, \varepsilon),$$

where V is a set of typed roles, E a set of typed directed or undirected relations, χ a set of invariants or constraints, γ a boundary/context record, q the quantity of interest, and ε the evidence identities supporting the extraction. Domain-facing entities are retained in provenance, but the transfer layer reasons over roles and relations.

The object is intentionally scoped. A hospital and a packet network need not be globally "the same." They may instantiate the same queue-stability substructure for a backlog or delay QoI while differing radically in ethics, priority policies, intervention constraints and entity semantics. The formalism therefore uses a directional mapping witness rather than a global equivalence relation.

Definition 1 (Directional structural witness)**.** *For source S_A and target S_B, a witness is*

$$w_{A\to B} = (\mu_V, I^+, I^-, B, E_w, U_w),$$

where μ_V maps source roles to target roles, I^+ records preserved invariants, I^- records explicitly non-preserved properties, B contains required target boundary conditions, E_w is evidence for the mapping and U_w records uncertainty. The witness is licensed only for the registered QoI.

The relation is directional because a transformation useful in A may require information or interventions unavailable in B. It is context-specific because a mapping that is valid in one regime can fail after a boundary change. It need not be transitive: $A \to B$ and $B \to C$ do not automatically license $A \to C$ unless the composed invariant and boundary obligations are separately satisfied.

2.1 Transfer gate

The reference transfer-gate module checks four non-compensatory conditions. First, the source and target endpoints must match the witnessed objects. Second, every required source relation must survive under the role mapping in the target. Third, the source invariants required for the intended operation must be declared and present in the target. Fourth, every required target boundary must match. QoI mismatch is itself a rejection. Domain labels and semantic similarity are not positive license conditions.

This gate is conservative by design. It can return `CANNOT_CHECK` for an invalid witness identity and `REJECTED` for an examined but failed mapping. A low semantic similarity is not a reason to reject when the relations and boundaries are witnessed; a high semantic similarity is not a reason to accept when the load-bearing invariant fails.

2.2 Why explicit non-preservation matters

Analogies are dangerous when the reusable part and the non-reusable part are stored in one undifferentiated similarity score. The witness therefore records I^-. In the queue example, "arrival competes with service and excess load grows backlog" may transfer while domain-specific priority policy and entity semantics do not. This is more than documentation: a later operator can declare which preserved invariants it depends on, and an intervention touching a non-preserved property can fail closed.

2.3 Operators as executable compressed knowledge

A later Orion layer can associate structural classes with operators

$$T = (\mathrm{Pre}, \mathrm{Transform}, \mathrm{Post}, B, C, F, E_T, A_T),$$

for preconditions, executable transformation, postconditions, boundaries, cost, failure modes, evidence and allowed authority effects. Paper 3 does not yet claim useful autonomous operator invention. The present objective is more basic: determine whether the structural object can safely retrieve or adapt an already validated operator across surface-disjoint domains. Operator induction becomes a later extension only after this transfer mechanism survives the benchmark.

2.4 Task-conditioned structural quotients: erasure must be licensed, not merely useful

A directional witness answers whether a structural object already extracted from one problem can be transported into another. A preceding problem is more basic: which distinctions in the source problem should enter that structural object at all? A raw representation may contain entity names, narrative order, coordinate choices, formatting conventions and other surface variables together with the relations that actually determine a registered quantity of interest. Treating all of these coordinates as equally load-bearing can make retrieval and transfer depend on irrelevant detail. Removing them indiscriminately is equally unsafe. We therefore introduce a *task-conditioned structural quotient* as a derived, evidence-bound computational view rather than as a replacement for the original problem.

Let P denote an immutable source problem representation, q a frozen quantity of interest and c a frozen context. A quotient proposal is a map

$$Q_{q,c} : P \longrightarrow Z,$$

where Z retains a registered set of structural coordinates and identifies a complementary set of candidate nuisance coordinates to erase or conditionally erase. The map is intentionally target dependent: a coordinate can be irrelevant for one QoI and indispensable for another.

Definition 2 (Task-conditioned structural quotient contract)**.** *A TCSQ proposal for* (P, q, c) *is a tuple*

$$\mathcal{Q} = (I^+, E^-, E^?, \Gamma, I_{\mathrm{inv}}, B, O, R, F, L),$$

where I^+ *are preserved coordinates,* E^- *are erased coordinates,* $E^?$ *are conditionally erased coordinates,* Γ *contains equivalence generators or declared nuisance transformations,* I_{inv} *are invariants that must survive the reduction,* B *are protected coordinates,* O *are sufficiency obligations,* R *records reconstruction bindings back to the source problem,* F *are falsifiers, and* L *are forbidden losses. The three coordinate sets* $I^+, E^-, E^?$ *must form a disjoint cover of the registered source coordinates, and every protected coordinate must lie in* I^+*.*

The proposal has no authority to certify its own sufficiency. A separately bound validation report assigns a verdict

$$V_Q \in \{\mathrm{VALID\_EXACT}, \mathrm{VALID\_APPROXIMATE}, \mathrm{REJECT}, \mathrm{CANNOT\_CHECK}\}.$$

Materialization for solver use is permitted only when every registered sufficiency obligation has been verified, no obligation has failed or remains unknown, protected-coordinate checks pass, evidence is content-bound to the exact proposal and source hashes, and at least one declared validation route has run. An approximate quotient additionally requires an explicit error metric and tolerance. Thus failure to establish irrelevance is not converted into permission to erase.

Solution sufficiency. For an exact quotient, the strongest ideal is that the target solution functional factors through the quotient on the declared scope:

$$F_{q,c}(P) = H_{q,c}(Q_{q,c}(P)).$$

The implementation does not infer this equality from compression ratio, retrieval performance or model confidence. Instead it records finite, typed obligations whose verification route can be domain specific: exhaustive oracle comparison on a known world, metamorphic tests over nuisance orbits, formal proof, counterexample search, or an externally governed scientific evaluation. For approximate quotients, the factorization is replaced by a registered bound such as

$$d(F_{q,c}(P), H_{q,c}(Q_{q,c}(P))) \leq \epsilon,$$

with the metric d and tolerance ϵ frozen before use.

Reconstruction and original-problem verification. A quotient-side result is never promoted merely because it is correct on Z. A reconstruction operator

$$R(s_Z; P, \mathcal{Q}) \longrightarrow s_P$$

restores the bindings required by the original problem, after which the ordinary original-problem verifier must accept s_P. This separates three claims that generic compression can easily blur:

$$\text{quotient valid} \neq \text{quotient solution valid} \neq \text{original problem solved.}$$

A reconstruction failure or failed original-problem verification therefore closes the route without rewriting the source problem.

Derived view rather than destructive simplification. The canonical source remains immutable. The current implementation materializes an accepted quotient as a derived memory view whose erasure ledger is explicit and whose authority certificates are copied from a pinned canonical source rather than caller-supplied. The same validated view may compile a derived problem fibre for retrieval; absence of a coordinate from that fibre means only that the coordinate is computationally suppressed for (q, c) under the registered certificate. It does not mean that the coordinate is false, scientifically absent, or irrelevant to another target.

This distinction allows TCSQ to compose with the directional transfer machinery. The intended pipeline is

$$P \xrightarrow{Q_{q,c}} Z \longrightarrow S_{q,c} \xrightarrow{w_{A\to B}} \text{candidate transfer},$$

where the quotient decides which source distinctions are admitted into the task-conditioned structural view and the later witness decides whether those retained relations may be transported to a target. Protected coordinates, preserved invariants and forbidden losses are carried into the obstruction/transport layer; erased coordinates are never silently reconstructed as transfer evidence.

2.4.1 Nearest work and the residual claim

Task-relevant compression itself is not new. The information-bottleneck principle formalizes compression subject to preservation of information about a relevant variable [22]; state-abstraction work studies representations that preserve action or value-relevant behavior [23, 24]; recent decision-sufficient representation work directly optimizes representations for downstream decisions [25]; and prompt-compression systems such as LongLLMLingua and LLMLingua-2 remove context tokens while attempting to preserve downstream model performance [26, 27]. Formal verification likewise has a long tradition of abstraction with counterexample-guided refinement rather than trusting a coarse abstraction unconditionally [28].

More recent parents narrow the residual further. Context Codec explicitly represents source-grounded semantic atoms, protects critical commitments, measures compression soundness/completeness, studies round-trip recoverability and recommends conservative fallback when low-confidence or safety-critical information would be lost [29]. Consequently, "auditable compression with protected information and verification" is not a TCSQ novelty claim. SLICEFORMER learns static program slices with dataflow-aware representations and constrained decoding [30], while STACK performs state-aware compression of reasoning traces to reduce redundant reasoning without discarding task-relevant knowledge [31]. These works also rule out novelty claims based merely on removing irrelevant program/context/reasoning state.

The candidate residual here is therefore narrower and problem-solving specific. TCSQ asks whether a scientific problem object can be quotiented under a frozen QoI/context by an explicit structural equivalence/erasure contract, then used as the input to the same problem-fibre and directional-transfer machinery while preserving source lineage, protected coordinates, invariants and forbidden losses. Solver-side sufficiency is separate from compression fidelity, and a quotient-side solution must be reconstructed and verified against the exact original problem before any scientific promotion. The quotient is also subject to the framework's authority non-escalation rule: computational omission cannot mint, revoke or upgrade scientific authority. In the stronger empirical programme, Context-Codec-style verifiable compression, generic prompt compression, state abstraction, structure-without-erasure and mechanism-alignment controls are therefore comparators rather than strawmen.

Whether these additional obligations improve total task cost, nuisance robustness or cross-domain retrieval remains an empirical question. The present implementation plus known-world SQ-1/SQ-2 studies establish a fail-closed contract and controlled sufficiency/dependency-recovery results; they do not establish that a language model can discover valid quotients, that TCSQ is minimal, or that it outperforms strong compression/abstraction baselines on natural research tasks.

3 Representation charts and solver portals reuse the same preservation discipline

The unified problem-solving projection introduces representation charts only as computational views. A candidate portal

$$F_{ij} : R_i \to R_j$$

can change which operators are applicable or sharply reduce search cost, but its admissibility is governed by the same directional, QoI- and boundary-scoped preservation logic as an ordinary structural transfer. The portal must state what is preserved, what is not preserved, the target scope, assumptions, reconstruction/round-trip obligations and the verifier used to validate the translation. A representation that makes a problem easy by silently deleting a protected coordinate is therefore a rejected transfer, not a useful "wormhole."

This gives the transformation-effect vocabulary—for example `PRECONDITION`, `CONTINUATION`, `QUOTIENT`, `DUALIZE`, `LOCALIZE`, `DECOUPLE` or compilation to a specialist solver—a precise structural role. These labels describe the intended computational effect and are routing metadata; they do not replace the witness. A Verified Solver Compilation candidate can be marked validated even for routing only after a separately bound preservation receipt exists. Multi-chart search then composes through ordinary GLUE/interface obligations, and any unresolved preservation condition returns `CANNOT_CHECK` rather than being compensated by a shorter predicted route.

The empirical question is consequently narrower than "do representation changes help?" The open residual is whether Orion can diagnose a needed transformation effect, locate or construct a valid portal, and reduce total verified solving cost beyond strong fixed-representation and solver-selection controls. That result is not established by the present structural-contract evidence.

3.1 Empirical evidence argues against symmetric structural equivalence

The directional witness is not only a conservative formal choice. Wang reports a controlled asymmetry in structural transfer between natural-language and biological foundation models [19]. Language models trained or fine-tuned on structural language tasks retain meaningful competence when moved to biological-sequence structure, while the reverse transfer remains weak across matched-token controls, scaling and multiple biological model families. The result is domain-specific and does not establish a universal law of transfer, but it is a direct counterexample to the assumption that shared structural regularity implies symmetric representational transfer.

For Paper 3 this has two consequences. First, a structural relation should default to a directed edge $S_A \to S_B$ with a witness rather than an undirected equivalence class. Second, transfer evaluation should include direction reversal as an explicit perturbation. If $A \to B$ succeeds but $B \to A$ fails, the library should preserve that asymmetry rather than canonicalize the two tasks into one globally interchangeable class. Any later transitive composition likewise requires a new scope/invariant check.

This also strengthens the Q2/Q3 benchmark logic. The object being tested is not whether two domains "look analogous" in a human narrative sense. It is whether a particular source-side operator or inference depends only on relations and invariants that are actually available in the target under the registered boundary conditions. Directional negative-transfer history is therefore part of the reusable state.

3.2 Executable transport obligations and fail-closed abstention

The strengthened implementation makes the witness more specific than a structural-similarity score. A directional witness is interpreted as a finite set of typed transport obligations

$$\Omega^{(q)}_{A \to B} = \{o_1, \ldots, o_m\},$$

scoped to quantity of interest q. An obligation records a role, relation, invariant, source precondition, target boundary, QoI requirement or forbidden loss together with its requirement class, evidence identities and a three-valued verification status

$$\text{status}(o_i) \in \{\text{SATISFIED}, \text{VIOLATED}, \text{UNKNOWN}\}.$$

The distinction between `VIOLATED` and `UNKNOWN` is load-bearing. Failure to establish a required precondition is not itself evidence that the precondition is false.

The transfer decision is deliberately non-compensatory. Write $\text{Transfer}(A \to B; q)$ for the directional decision under QoI q. Then:

- LICENSED if every load-bearing obligation is satisfied;
- REJECTED if at least one load-bearing obligation is demonstrably violated;
- CANNOT_CHECK otherwise, whenever at least one load-bearing obligation remains unresolved.

Thus several preserved relations cannot compensate for a violated target boundary, wrong transfer direction or forbidden loss. Conversely, a property declared irrelevant to the registered QoI may be listed as a permissible loss rather than forcing global structural equivalence. Each load-bearing obligation is content-bound to evidence rather than inheriting authority merely

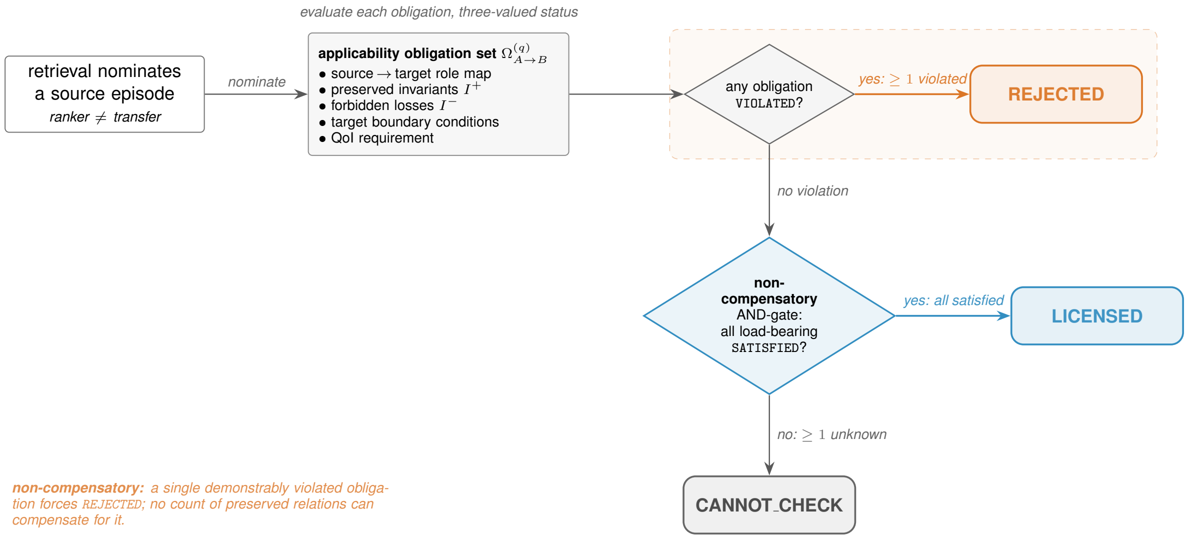

Figure 1: The fail-closed transfer decision. A nominated source is evaluated against the applicability obligation set (role map, preserved invariants I^+, forbidden losses I^-, target boundary, QoI), each with a three-valued status. The rule is non-compensatory: any single demonstrably violated load-bearing obligation forces `REJECTED`; only when every load-bearing obligation is satisfied is the transfer `LICENSED`; an unresolved obligation yields `CANNOT_CHECK`. No count of preserved relations can buy back a violation.

because the witness object contains some evidence somewhere. The implementation emits an obligation-level trace and a deterministic content hash so that a confirmatory benchmark can freeze the exact witness contract before outcome access. The full fail-closed decision procedure is shown in Figure 1.

These semantics are an implementation and evaluation contract, not an empirical superiority result. The confirmatory question remains whether machine-produced obligations improve valid distant transfer and reduce hostile semantic near-misses relative to strong semantic, relational and mechanism-alignment controls under matched resources.

4 Directional witnesses as Orion experience-transfer relations

Orion makes experience persistence an implemented architectural fact but leaves transfer validity as an evidence question. A source `TaskEpisode` says what happened in one registered context. A versioned `Lesson` is a scoped abstraction over one or more episodes. Neither object is automatically applicable to a new target. The role of the directional structural witness is to supply an explicit candidate relation

$$W^{(q)}_{s\to t} : (E_s, L_s, \gamma_s) \rightsquigarrow (P_t, q, \gamma_t)$$

whose obligations state which roles and invariants are preserved, which properties are not preserved, and which target boundaries would invalidate reuse.

This makes the witness directional in two senses. First, the target quantity of interest determines which source structure is load-bearing. Second, a valid mapping from source to target need not license the reverse mapping because scope, information loss or boundary conditions can be asymmetric. A witness is therefore not an embedding similarity score and not an equivalence class over tasks.

4.1 Transfer can alter search priority without creating authority

When a witness passes the registered structural and boundary gate, v3 may use the linked episode or lesson to increase retrieval priority, activate an already promoted tool, or place an operator path earlier in the search queue. The permitted inference is

$$\text{witnessed applicability} \Rightarrow \text{eligible experience reuse},$$

not

$$\text{witnessed applicability} \Rightarrow \text{scientific claim true}.$$

Scientific authority remains owned by the target problem's verification path. This is the cross-domain specialization of the framework invariant that experience-conditioned routing is not epistemic authority.

4.2 The witness is the transport obligation inside JUMP and GLUE

The deployed obstruction–transformation router now gives the witness a precise operational location. The router first fingerprints the active target obstruction using roles, relations, constraints, failure mechanisms, invariants to preserve, desired transition and forbidden losses. It then queries a content-bound memory of recorded source episodes

$$O_s \xrightarrow{T_s} O'_s.$$

A semantic or structural ranker may nominate a source, but source retrieval is not yet transfer.

For a cross-domain `JUMP`, the source transformation becomes a viable target proposal only when a mapping witness accounts for the load-bearing source structure. In the executable contract this includes source-to-target role mapping, shared relations and constraints, every enabling source precondition, material disanalogies and target-domain validation obligations. An unrepaired source precondition or a conflict with a forbidden target invariant blocks the transfer even when the source event itself is well verified.

The same discipline matters for `GLUE`. A source episode whose recorded effects cover only part of the target transition is not promoted to a complete route merely because it is structurally similar. Partial episodes remain composition candidates. Their combined recorded effects must cover the desired transition, and a separate composition witness must bind operation order, interfaces, incompatibility checks and target validation. Directional witnesses therefore license transport of components; they do not by themselves prove that the composed target pipeline is coherent.

This yields a three-stage distinction:

$$\text{retrieved source} \neq \text{witnessed target applicability} \neq \text{verified target success}.$$

This paper studies the middle relation (Figure 2).

4.3 Negative transfer is learned through an evidence ladder

A failed transfer attempt first creates a non-success task episode. It does not immediately establish that the witness class, operator or source lesson is invalid. The system records the failure as observed-only, then permits competing diagnoses such as context mismatch, broken invariant, implementation defect, missing evidence or an unrelated downstream failure. Only discriminating evidence can promote a supported diagnosis, and only fresh replay/transfer can turn that diagnosis into a reusable boundary lesson.

The resulting discipline prevents one near-miss from becoming a global blacklist while still allowing repeated boundary-sensitive failures to improve later routing. It also makes negative-transfer history addressable: a future witness can explicitly inherit a known non-preserved property or target boundary rather than merely receiving a lower opaque score.

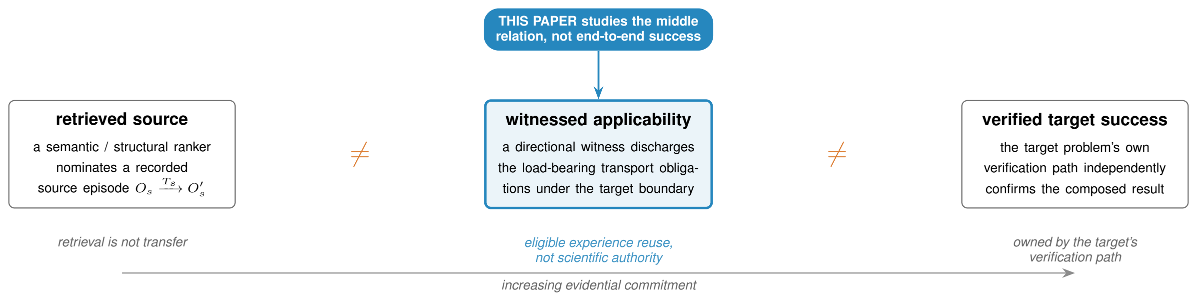

Figure 2: Three stages of increasing evidential commitment. A retriever nominating a source episode is not the same as a directional witness discharging the load-bearing transport obligations under the target boundary, which is in turn not the same as the target problem's own verification path independently confirming success. This paper's contribution is the middle relation—witnessed applicability—which licenses eligible reuse without minting scientific authority; retrieval and end-to-end success are owned by other stages.

Repeated transfer residuals also interact with the router's invention gate. They may contribute to a `LIFT` decision only after bounded direct, cross-domain and compositional search has been accounted for and the same residual feature appears across at least two distinct failures. Even then, the result is a missing-transformation specification rather than a new witnessed skill. The invented representation or operator must later earn target evidence separately.

4.4 Connection to Orion-triviality

V3 classifies a verified solution as `TRANSFER_NOVEL` when no new primitive was invented but pre-existing problem-solving structure was transported into a new context by an explicit mapping witness. The directional witness studied here is a candidate evidence object for that classification. A solution should receive `TRANSFER_NOVEL` only when the transfer relation is explicit, the target solution is independently verified, and resource ancestry shows that no new representation/operator/ontology was required.

This gives Paper II a narrower post-v3 claim. The framework already possesses a persistent experience substrate; this paper does not claim novelty for persistence itself. It asks whether one scientifically scoped, boundary-aware relation can make cross-domain reuse safer and more measurable than semantic resemblance alone. The natural-domain, strong-control and independent-annotation gates remain necessary before that question can receive an independent empirical answer. The new semantic-shortcut implementation therefore strengthens the role and testability of the witness without converting the existing demoted pilot evidence into confirmatory natural-domain validation.

5 A semantic–structural factorial benchmark

What is established versus what is registered. To read the evidence correctly, three lanes must be kept distinct. The **objective known-world lane** (Section 5.7) is *complete*: a preregistered $n = 576$ exact-verifier study in which the full applicability contract matched every verifier decision. The **natural-domain lane** (this section and Section 5.9) is a *power-limited, demoted* diagnostic ($n = 16$), blocked on independent annotation and reported as such, not as a confirmatory result. The **cost, learning and reporting programme** (Sections 6–8) is a *preregistered plan* for a future empirical study, not a result of this paper. Everything below in this section is diagnostic or planning; the confirmed result is in Section 5.7. Figure 3 summarizes the status of each lane.

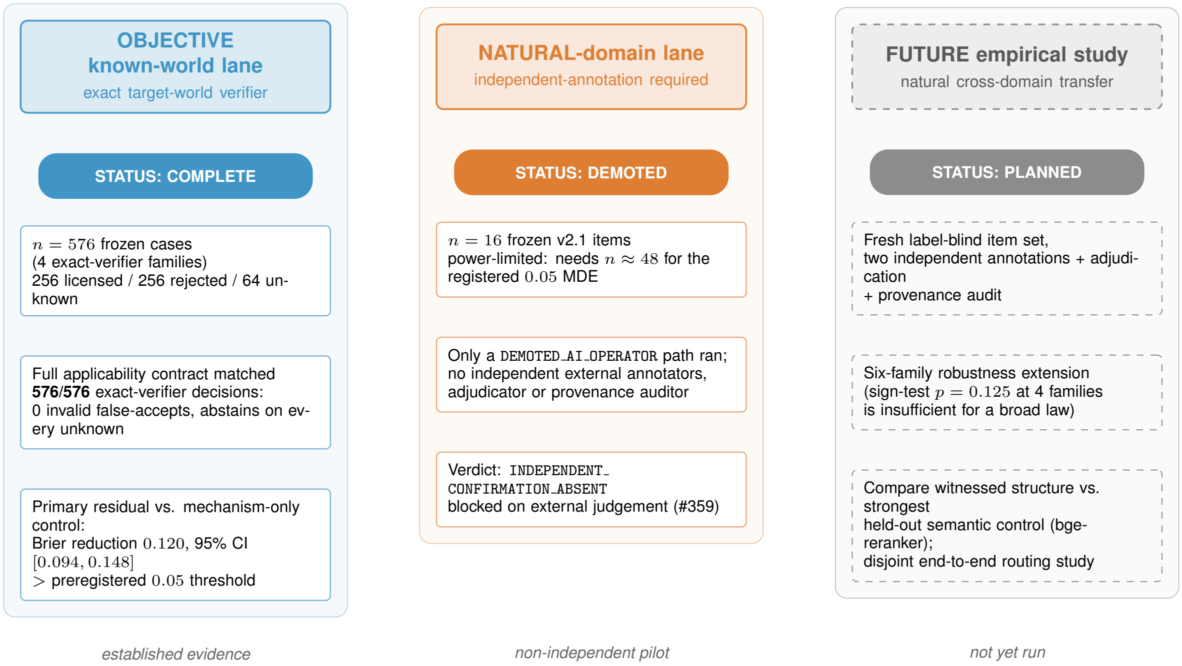

Figure 3: The three evidence lanes and their status. The objective known-world lane is a completed, preregistered $n = 576$ exact-verifier study in which the full applicability contract matched all 576 verifier decisions. The natural-domain lane is a power-limited, demoted pilot ($n = 16$) with no independent annotation. The future empirical study is preregistered but not yet run. Only the objective lane is established evidence; the others are explicitly a non-independent pilot and a plan.

The central benchmark independently manipulates surface-semantic similarity and structural similarity. Let $S_{\text{sem}} \in \{\text{low}, \text{high}\}$ and $S_{\text{str}} \in \{\text{low}, \text{high}\}$. The four cells are:

Cell	Semantic relation	Structural relation
Q1	high	high
Q2	low	high
Q3	high	low
Q4	low	low

Q2 and Q3 are the load-bearing cells. Q2 asks whether the system can transfer when vocabulary and domain semantics are deliberately distant but the relevant relation/invariant structure is preserved. Q3 is a semantic decoy: vocabulary is similar and embedding similarity should be high, but the causal/dynamical structure or boundary condition that justifies transfer is changed. A useful structural system should accept Q2 and reject Q3.

The deterministic benchmark scaffold now exercises this requirement across three distinct mechanism families rather than one illustrative analogy. For *queue stability*, a computer network and an emergency department use different entity vocabularies but share a continual-arrival/finite-service/backlog structure for the registered QoI; the high-semantic decoy instead uses queue language in a finite-batch regime, so the steady-flow stability invariant is invalid. For *positive feedback*, viral adoption and a bank-run withdrawal loop share the sign structure required for perturbation amplification, while a semantically similar control-feedback target reverses one loop edge and becomes stabilizing negative feedback. For a *threshold cascade*, epidemic spread and cascading grid outage share a supercritical propagation structure, while a same-domain subcritical cascade decoy removes the load-bearing reproduction relation and violates the threshold/connectivity boundary. The original Q4 case remains a semantically and structurally unrelated queue-to-feedback mapping.

5.1 Deterministic conformance result

The exact-subject executable receipt contains six load-bearing Q2/Q3 cases: one Q2 and one Q3 case in each of the three mechanism families. The deliberately semantic-only decision rule licenses high-semantic candidates and rejects low-semantic candidates; it is therefore wrong on all six hard cases by construction. The structural gate instead checks the registered directional witness, QoI, load-bearing relations, invariants and target boundaries. It licenses all three Q2 transfers and rejects all three Q3 decoys.

Decision rule	Q2 valid licensed	Q3 invalid accepted	Hard cases correct
Semantic-only control	0/3	3/3	0/6
Structural witness gate	3/3	0/3	6/6

This table is a software-conformance result, not a statistical generalization result. The cases were deliberately constructed to separate semantic and structural similarity; their role is to establish that the implementation can express the intended distinction before model-level experiments are authorized. They provide no estimate of performance on naturally occurring cross-domain mappings.

These cases are deterministic conformance worlds, not an empirical claim that the representation generalizes. Their purpose is to prevent a one-family implementation from earning an expensive training run merely by memorizing queue-specific details. The confirmatory dataset will still require independent annotations and broader families such as flow conservation, bottlenecks, delayed negative feedback, diffusion/contagion, matching, scheduling, concentration fragility, path dependence and search/backtracking. Each family should span multiple surface domains and include adversarial near-misses.

5.2 A broader internal diagnostic fails closed at the annotation gate

Before inspecting predictor outcomes, we froze a balanced Q1–Q4 proposal set, leave-one-family-out split, predictor family, metrics and stopping thresholds at Git subject `f2701f732f83`. The resulting diagnostic contains 44 constructed proposal pairs across 11 proposed mechanism families. Every coordinate was proposed in the same assistant session that assembled the benchmark. None has two independent human/expert annotations or adjudication, so all 44 are explicitly marked ineligible for confirmatory use.

The local pilot fits deterministic L2-regularized logistic predictors using successively richer feature sets: surface overlap, skill-tag overlap, dependency-pattern overlap and witnessed structure. It is a cheap predictive diagnostic with no foundation-model judgement. Leave-one-family-out outcomes crossed the frozen incremental-signal thresholds: relative to the strongest of these frozen lexical/tag controls, the dependency-aware model, witnessed structure increased ROC-AUC from 0.914 to 1.000 and average precision from 0.903 to 1.000, while reducing Brier loss from 0.166 to 0.039. Its Q2 true-accept rate was 1.00 and Q3 false-accept rate was 0.00. These point estimates measure discrimination of labels encoded in an internally constructed proposal set; they do not estimate human agreement, natural transfer performance or incremental value beyond a modern content encoder.

The independent-annotation gate nevertheless has authority over the diagnostic-signal gate. Its exact verdict is `FAIL_CLOSED_MISSING_INDEPENDENT_ANNOTATION`: zero of 44 items are confirmatory-eligible. Consequently, no training or inference run was launched. The unresolved residual is evidence acquisition, not a license to substitute same-session agreement or spend compute defending the original shared-substrate headline.

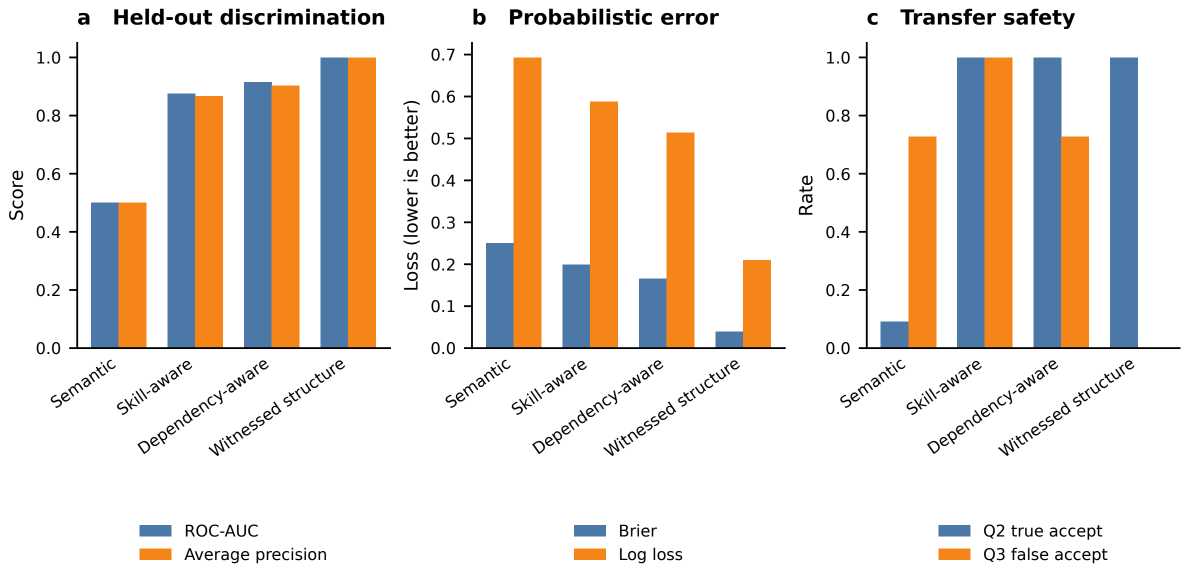

Figure 4: Receipt-bound internal cheap-gate diagnostic. The unit is a constructed source–target pair ($n = 44$) and each prediction is out-of-family under a deterministic leave-one-family-out fit. Bars are aggregate point estimates; there are no repeated seeds, sampling intervals or independent annotations. Panel a reports ROC-AUC and average precision, panel b Brier and log loss, and panel c Q2 true accept with Q3 false accept. The figure is generated only from the checked machine-readable result receipt; values are not hand-entered into the plot.

A subsequent chronology audit identified a stricter cut: the frozen v1 protocol records both that labels were visible during construction and that confirmatory use is not permitted. The 44-item v1 set is therefore permanently an internal constructed diagnostic; later annotations cannot retroactively convert it into confirmatory evidence or authorize compute. We retain that receipt unchanged as negative history and register a separate v2 lane. V2 requires a fresh label-blind item set frozen before labels or results, the same or stricter signal thresholds, two genuinely independent human/expert annotations per item, distinct post-freeze adjudication and an external provenance audit. Repository validation can check hashes, coverage, chronology and attestations, but cannot itself prove that an independence attestation is truthful. Until both the new annotation gate and held-out signal gate pass, the LUNARC preflight refuses training and inference submission.

The first v2 curation tranche now freezes 16 fresh label-blind source–target descriptions across four previously unused families: bottleneck–minimum-cut, diffusive transport, matching and allocation, and path dependence and reinforcement. The artifact binds 19 citation records and preserves four items per family, but it contains zero external judgements, no adjudication and no signal estimate. Its role is chronological: it fixes what future annotators will assess before their answers exist. Consequently no training or inference job is authorized; the writable FS9 workspace and GPU-account observation are execution preparation only.

A pre-annotation identifier-leakage audit then found that four curator-only source identifiers included the phrase `near_miss`. Although those identifiers were not outcome fields, they could reveal a structural expectation if an annotator inspected the source artifact or reconstructed the linkage. We therefore preserve that first source set as negative history and replace it with an otherwise byte-equivalent v2.1 set whose identifiers are neutral sequential codes. The earlier packet attempt was discarded and its private linkage removed. This repair was completed with zero external judgements and was superseded before any external judgement; no training or inference job is authorized.

5.3 Structural score must add information beyond semantic score

For a transfer outcome Y, the confirmatory analysis compares models such as

$$\operatorname{logit} P(Y=1) = \beta_0 + \beta_1 S_{\text{sem}}$$

against

$$\operatorname{logit} P(Y=1) = \beta_0 + \beta_1 S_{\text{sem}} + \beta_2 S_{\text{str}} + \beta_3 S_{\text{sem}} S_{\text{str}} + u_{\text{family}}.$$

The central mechanism requires material incremental predictive value from the structural term and, operationally, a low invalid-transfer rate in Q3. If a strong semantic retriever or LLM analogy judge already predicts valid transfer as well as the witnessed structure, the proposed representation does not earn an additional layer.

The frozen confirmatory item set holds 16 items, and at that size a bare point-estimate threshold cannot distinguish signal from sampling noise. The Hanley–McNeil paired standard error for an AUC difference over 16 items ranges from 0.067 to 0.176 across plausible inter-arm correlation and label-balance assumptions, so a frozen gain threshold near 0.05 sits within one third of a standard error of pure noise and returns the wrong verdict roughly one time in three, in both the false-pass and false-fail directions. The Q2 and Q3 acceptance rates are proportions over approximately eight items each, where one item moves either rate by 0.125 and the 95% interval half-width at $p = 0.5$ is ± 0.35 against thresholds of 0.8 and 0.2. A point estimate alone is therefore uninterpretable at this sample size.

The confirmatory gate consequently carries an uncertainty-quantification layer that the protocol froze before any annotation arrived. For each item the per-item Brier score is computed for the witnessed-structure arm and the strongest control arm under the shared adjudicated label, and the paired difference (control minus structural) forms the resampling unit. A paired bootstrap confidence interval and a sign-flip permutation p-value, both implemented in a standard-library inference helper, are evaluated with a frozen seed and 10,000 resamples. The diagnostic gate passes only when this interval excludes zero in addition to every frozen point-estimate threshold; a point estimate that clears 0.05 while its interval overlaps zero is reported as indistinguishable from noise rather than as confirmation. This is an evaluation-design repair: it changes what the gate can say about a future annotated outcome, and no result is claimed, supported or refuted here.

5.4 Annotation and leakage

A human-curated version will record role graphs, typed relations, invariants, boundary conditions and QoI with at least two independent annotations plus adjudication. Inter-annotator agreement is reported separately for roles, relations, class assignment and boundaries. The final domain split must prevent a surface domain from appearing in both training and target evaluation for the same structure family. Synthetic generation can expand coverage, but held-out human adjudication is required to show that the structural labels are not merely artifacts of the generator.

5.5 Fresh v2.1 annotation packet

After the neutral-identifier repair merged, we generated a new packet from the exact v2.1 source set and bound it to merged Git subject `f4cee8313ec64d02873b87f92c51c35c113cd70d`. The public packet contains 16 opaque item identifiers and no curator source identifiers, outcomes, annotations, adjudication or signal result. A coordinator-only linkage is held outside Git on FS9 with mode `0600`; the public receipt records only its location and content hashes. The discarded pre-repair packet was not reused.

A same-context hostile usability check found that the original response schemas forced `cannot_assess=false`, despite the frozen rubric instructing annotators not to guess under missing evidence. The repaired schemas permit `cannot_assess=true` only when all nine judgement coordinates are null. The compiler preserves that submission but fails confirmatory import, so missing evidence remains visible negative history rather than a coerced label. No external judgement exists at this packet-freeze stage, and training and inference remain unauthorized.

5.6 A content-bound semantic control is frozen before labels

The v2 evaluator audit exposed a weak-control residual: surface Jaccard, skill-tag Jaccard and dependency-tag Jaccard are transparent conformance controls, not a strongest-feasible modern semantic comparator. Before any external judgement became visible, we therefore froze a separate content-bound control using the exact revision `953dc6f6f85a1b2d` of `BAAI/bge-reranker-v2-m3`. Its input is a deterministic source/target rendering of the v2.1 domain, surface terms, skill tags, dependencies, citation/title strings and shared QoI. Every rendered side and pair receives a SHA-256 binding. Candidate invariant/boundary proposals, annotations, quadrants and transfer outcomes are excluded, so the control cannot silently consume the structural target it is meant to challenge.

This is a protocol freeze rather than a result. An earlier clean-environment descriptor attempt returned `CANNOT_CHECK_MODEL_ASSET_MISSING` because the 2.27-GB model artifact was absent from that checkout and emitted zero scores. Subsequent FS9 harvests (jobs 3476527–3476529) reached `HARVEST_DESCRIPTOR_READY` with sixteen label-blind records while keeping `training_authorized=false`. At that descriptor-harvest freeze, repository issue `#217` still recorded zero public independent annotator, adjudicator or provenance-auditor responses, so the zero-label chronology observation remained in force; #217 was later closed on a demoted `AI_OPERATOR / NON_INDEPENDENT` path, with the independent-human residual continuing under open successor #359. Descriptor readiness remains execution preparation only: it does not mint labels, authorize training or inference, or support confirmatory Paper III metrics. The eventual signal test must still await external annotations and adjudication, and must compare witnessed structure against the strongest held-out non-structural arm including the reranker and its skill/dependency augmentation; selecting the older lexical control after seeing labels is forbidden.

5.7 Preregistered objective known-world confirmatory result

The historical Q1–Q4 diagnostic uses internally proposed cases and therefore cannot by itself establish an external transfer-validity construct. We consequently registered a separate objective lane in which the correct transfer decision is obtained by executing a family-specific target-world verifier rather than by an annotator or by the generator's perturbation identity. The primary v1 scope was frozen before confirmatory outcome access to four heterogeneous exact-verifier families: mapped directed-flow feasibility, Horn-style logical entailment, dimensional/unit transformations and finite deterministic state-transition systems. The first development instrument failed the preregistered coordinate-ablated-twin circularity attack and was preserved as negative history rather than promoted. A redesigned development instrument passed the registered semantic-difficulty and circularity checks; its paired binary-Brier variance gave a required decidable sample of 431 for the registered 0.05 MDE. Before generating confirmatory outcomes we therefore froze seed `2026081202`, 16 instances per base cell and $n = 576$ total cases. The freeze was merged to the public repository before that seed was executed.

The fresh confirmatory packet contained 256 verifier-licensed transfers, 256 verifier-rejected transfers and 64 `CANNOT_CHECK` cases. A lexical/Jaccard control remained near chance on the 512 decidable cases (accuracy 0.496). More importantly, the strongest deterministic structural control in this lane was not lexical: a derived-effect or "mechanism-only" rule already reached exact three-way accuracy 0.875, accepted every valid transfer and rejected every registered semantic-near-miss case, but still licensed 25% of all verifier-invalid transfers because mechanism preservation alone did not enforce every applicability coordinate. A relation/invariant-only

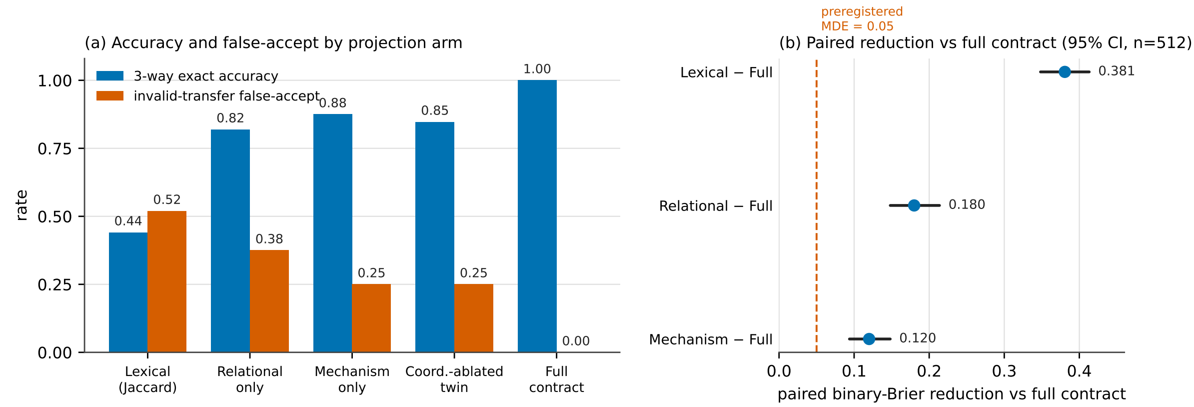

Figure 5: Preregistered objective known-world confirmatory result (four families, $n = 576$; 512 decidable cases). **(a)** Three-way exact accuracy and invalid-transfer false-accept rate for each projection arm, ordered by increasing applicability structure. The mechanism-only control already reaches 0.875 exact accuracy yet still licenses 25% of verifier-invalid transfers; only the full applicability contract attains exact accuracy 1.0 with zero invalid false-accepts. **(b)** Paired binary-Brier reduction of each control relative to the full contract with item-bootstrap 95% confidence intervals (20,000 resamples); every reduction exceeds the preregistered 0.05 material-effect threshold (dashed). Values are read directly from the committed frozen confirmatory evidence; the figure visualizes the primary four-family outcome only and does not use the pending six-family robustness extension. Because the full arm computes its coordinates from machine-readable target-world facts, the result establishes *which* applicability coordinates a fail-closed gate must enforce, not that a system recovers them from unstructured prose.

control reached exact accuracy 0.819 with invalid-transfer false-accept 0.375. The full applicability contract—QoI, boundary, mapping, relations, source preconditions/invariants and independently derived target effect—matched all 576 exact-verifier decisions, retaining every valid transfer, falsely licensing no invalid transfer and abstaining on every verifier-unknown case.

The preregistered primary residual is therefore the full contract relative to the mechanism-only control rather than the much easier lexical comparison. On the 512 decidable cases, the paired binary-Brier reduction was 0.120 with an item-bootstrap 95% interval $[0.0938, 0.1481]$ from 20,000 resamples, exceeding the registered 0.05 material-effect threshold (Figure 5). The full-minus-mechanism exact-accuracy residual was positive in each of the four primary families.

Four family clusters are nevertheless insufficient for a broad cross-domain law: the two-sided sign-test value for four positive family signs is $p = 0.125$. A six-family robustness extension was named and frozen as a separate future coordinate before the primary outcomes; it is required before we claim broad family generalization.

This result is deliberately narrower than a model-level transfer claim. Candidate inputs in the objective lane expose machine-readable target-world facts from which the deterministic witness coordinates can be computed. The experiment therefore establishes that, in this generated exact-verifier scope, explicit applicability information resolves transfer decisions that remain ambiguous under relation-only and mechanism-only projections. It does not show that a language model can recover the witness from raw scientific prose, that a frontier direct-judgement model is inferior, that the same effect holds in natural scientific domains, or that witness gating improves end-to-end research outcomes. Strong semantic/LLM comparators, the disjoint downstream-routing study and independent natural-domain judgement remain separate evidence coordinates.

All four confirmatory packets are byte-reproducible from the committed generator, frozen seed and canonical JSONL serialization. Their expected SHA-256 identities, the paired inference, family analysis, anti-degeneracy audit and exact claim boundary are frozen in the committed

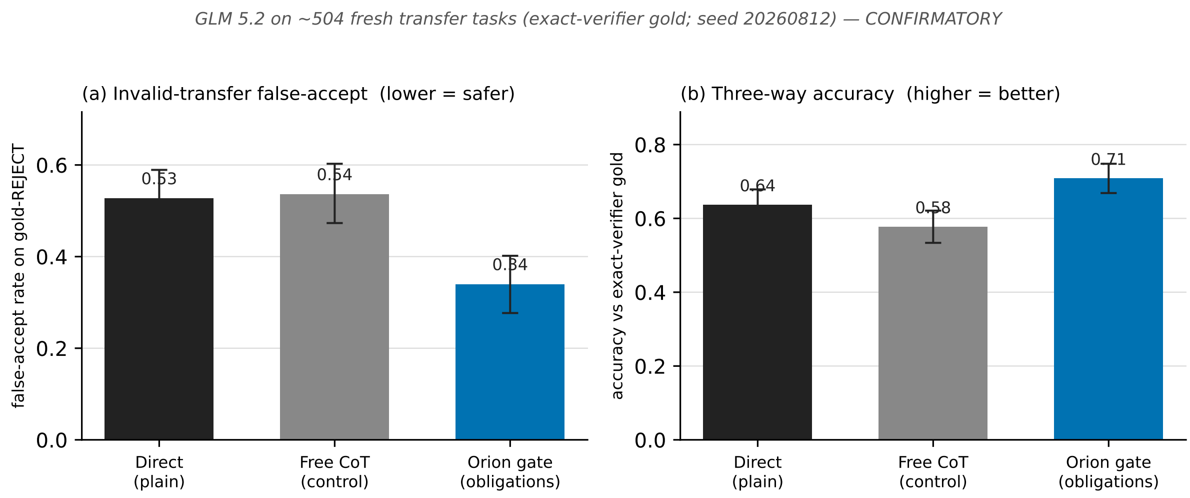

Figure 6: Preliminary external-model applicability-gate comparator (one fixed frontier model; $n \approx 504$ fresh exact-verifier tasks; bootstrap 95% CIs, 20,000 resamples). **(a)** Driving the model through the fail-closed applicability gate cuts the invalid-transfer false-accept rate from $0.527\,[0.460, 0.589]$ (Direct) to $0.339\,[0.277, 0.402]$ (gate); the two intervals do not overlap. The compute-matched Free-CoT control ($0.536\,[0.473, 0.603]$) is statistically indistinguishable from Direct, so the reduction is attributable to the obligation *structure*, not to additional reasoning. **(b)** Three-way accuracy against exact-verifier gold rises in step ($0.637 \to 0.708$). This is a system/method effect (a structured-inference scaffold), not a claim that the model reasons better; on the hardest semantic near-miss decoys the gate does not beat Direct, a real and reported weakness.

confirmatory evidence packet.

5.8 Preliminary external-model applicability-gate comparator

The confirmatory lane above isolates *which* applicability coordinates decide transfer, but it computes those coordinates from machine-readable target facts rather than asking a language model to apply them. We therefore ran a separate, independently frozen comparator that closes exactly this gap: it drives one fixed frontier model through the fail-closed applicability contract and measures whether the *obligation structure*—not extra reasoning tokens—reduces invalid-transfer false-accepts. The design (three conditions, primary metric, hypothesis direction and stopping rule) was frozen before any model output was scored; a fresh benchmark seed (`20260812`) and $n \approx 504$ exact-verifier tasks ($\geq$ the registered $n \approx 431$ decidable target) were used, with the same three-way label space in every arm. The three conditions differ only in the instruction: a plain Direct judgement, a compute-matched Free-CoT control ("think step by step"), and the Orion gate (check each named applicability obligation; accept only if all load-bearing obligations are satisfied; reject on any violation; abstain if unverifiable).

Figure 6 shows the gate roughly halving the odds of licensing a verifier-invalid transfer while the compute-matched control moves nothing, which is the isolation the design was built to obtain: fail-closed applicability obligations, not more thinking, carry the reduction. We report this as a *preliminary* comparator, deliberately outside the frozen four-family confirmatory packet governance above and subject to the same broadening the primary lane requires: it is one model, one seed and one benchmark family-set, it measures a scaffold rather than a smarter model, and it does not help on the trickiest semantic near-misses. A cross-model, cross-seed replication with the six-family sign test is the named next coordinate before any broad external-model claim.

5.9 Current confirmatory-data status

The strong semantic-control execution lane has advanced beyond the original manuscript freeze. The exact `BAAI/bge-reranker-v2-m3` asset was staged on FS9 and the label-blind descriptor jobs 3476527–3476529 reached `HARVEST_DESCRIPTOR_READY`. Sixteen v2.1 source–target items therefore have content-bound semantic descriptors under the preserved pre-annotation chronology.

The later operator-override path must be distinguished from independent confirmation. Issue #217 was closed after a demoted `AI_OPERATOR/NON_INDEPENDENT` annotation path was added. That path does not attest independent external annotators, an independent adjudicator or an independent provenance auditor, and it does not grant Constitution-grade confirmatory authority. The first demoted LUNARC train/inference pair, 3476750/3476751, failed because the system Python environment lacked `torch`; those failures remain negative history. After the runner was hardened, demoted jobs 3476753 (training) and 3476754 (inference) passed under `DEMOTED_AI_OPERATOR` authority. They establish that the demoted execution path can run; they do not establish independent natural-domain structural signal.

The independent-human coordinate was subsequently frozen in issue #332 as `CANNOT_OBTAIN_INDEPENDENT_EXTERNAL_HUMANS` with verdict `DEMOTED_AI_OPERATOR_PATH_COMPLETE__INDEPENDENT_REVIEW_ABSENT`. The remaining human-only residual is now tracked by open successor issue #359, which still requires two genuinely external independent annotators, a distinct adjudicator and a distinct provenance auditor. AI-operator outputs are not promoted to those roles.

Consequently the independent confirmatory question remains unanswered: whether witnessed structural features add predictive value beyond the strongest frozen non-structural control on independently judged natural-domain cases. The repository contains a demoted pilot execution, not independent confirmation. Until qualifying external evidence exists, the correct scientific status is `INDEPENDENT_CONFIRMATION_ABSENT`; any demoted pilot metric must be labeled as non-independent and cannot license the paper's strongest natural-domain transfer claim.

5.10 Pre-label power design

Before any external judgement, a label-blind power study (#248) registered a primary paired Brier-reduction MDE of 0.05 and simulated paired bootstrap/sign-flip resolution under plausible item-level noise. The frozen v2.1 sixteen-item packet is *underpowered* for that MDE: adequate resolution requires roughly $n \approx 48$ items across the registered noise envelope, whereas the solicited confirmatory set remains $n = 16$. The study therefore scopes external validation as `CONFIRMATORY\_PACKET\_POWER\_LIMITED`: indistinguishable outcomes are inconclusive rather than negative evidence, and confirmatory refutation claims remain blocked when intervals include zero.

The power limitation is not repaired by the demoted AI-operator path. Even if a future independent-human response is obtained for the frozen $n = 16$ packet, the registered precision/MDE interpretation remains in force; the item set cannot be enlarged post-label and then represented as the same confirmatory epoch.

6 Structural amortization as a cost-to-capability hypothesis

Reuse is not free. Let C_I be the one-time cost of inducing, validating and storing a reusable structure. Let c_B be the baseline per-task cost of solving without reuse and c_R the per-task cost of retrieving, adapting and verifying the reusable structure. Then

$$C_B(n) = nc_B, \qquad C_R(n) = C_I + nc_R.$$

When $c_R < c_B$, reuse becomes strictly cheaper once

$$n > \frac{C_I}{c_B - c_R}.$$

The executable reference implementation reports this break-even count and returns no finite break-even when the reuse path is not cheaper per task. This elementary accounting is important because prior workflow-compilation work can reduce runtime calls without establishing net efficiency when offline compilation cost is omitted [11].

The empirical paper uses a fuller vector

$$C_{\text{total}} = C_{\text{induction}} + C_{\text{training}} + C_{\text{retrieval}} + C_{\text{adapt/reason}} + C_{\text{tools}} + C_{\text{verification}}.$$

No coordinate is silently set to zero. A cost-to-capability point is admissible only if it reaches the frozen capability target without crossing the registered validity-failure ceiling.

Hypothesis 1 (Structural transfer). *At matched semantic similarity, a witnessed structural score provides material incremental prediction of held-out transfer success.*

Hypothesis 2 (Semantic-decoy rejection). *A witness-aware transfer gate has a lower invalid-transfer rate than semantic retrieval or analogy-only baselines on Q3 cases with high semantic and low structural similarity.*

Hypothesis 3 (Static structural training efficiency). *At a frozen structural-OOD capability target, static structural curation reaches the target with fewer training tokens or examples than strong semantic/diversity and skill-graph parents in regimes containing cross-domain structural redundancy.*

Hypothesis 4 (Inference efficiency). *For a frozen base model and target capability, reuse of validated structural objects/operators reduces test-time reasoning/search cost relative to semantic memory, analogy prompting and strong skill/workflow-prior parents after all retrieval, adaptation and verification costs are included.*

Hypothesis 5 (Redundancy mechanism). *Under a controlled data generator, the advantage of structural reuse increases with structural redundancy after surface-semantic diversity is held approximately fixed.*

Hypothesis 6 (Shared substrate). *A structure identifier induced or registered during training/data curation can be reused at inference without changing to an unrelated task-specific representation. The strongest paper claim requires this shared object; two independent tricks would support two narrower results rather than one unification.*

6.1 Optional training-time extension: learner-conditioned structural saturation

The static training hypothesis above does not by itself imply that a structural family should receive the same training allocation throughout learning. Recent work already establishes substantial prior art for model-aware or online data selection, including evolving influence-based selection, online distribution optimization, model-state-dependent perplexity schedules and explicit missing-skill targeting [3, 4, 5, 6]. The present paper therefore claims no novelty for adaptive curricula, model-aware selection, saturation-aware reweighting, skill graphs or missing-skill profiles in general.

A narrower Orion extension remains open because the transfer substrate is not a flat skill label. For a registered structural object S, a future learner-conditioned training view may estimate a scoped mastery vector

$$m_t(S) = (m_P, m_C, m_B, m_R, m_T, m_F),$$

where the coordinates denote principle acquisition, composition mastery, boundary robustness, representation invariance, cross-domain transfer and retention/forgetting resistance. The vector is learner-specific: it is bound to a model/checkpoint and a frozen probe family, and it can become stale after later parameter updates. It is not scientific authority.

The load-bearing distinction is that one coordinate may saturate while another remains unsaturated. In particular,

$$\mathrm{mastered}(A) + \mathrm{mastered}(B) \not\Rightarrow \mathrm{mastered}(A + B),$$

and high in-domain principle accuracy does not establish boundary or cross-domain transfer mastery. This prevents "already knows the skill" from becoming a scalar excuse to remove all further examples containing that principle.

The training-time extension should therefore be interpreted as budget reallocation rather than destructive deduplication. Repeated exposure may still be useful for robustness, calibration, representation invariance and forgetting prevention; prior LLM curation work has shown that intelligently repeated data can outperform random single-use exposure [8]. A learner-conditioned scheduler must preserve a registered repetition/retention floor and preferentially move only the *marginal* budget toward unsaturated compositions, boundaries, representations, domains or hostile near-misses.

Hypothesis 7 (Learner-conditioned structural allocation — extension hypothesis)**.** *In a training distribution with surface-diverse structural redundancy, an adaptive scheduler conditioned on a frozen learner-specific mastery vector over registered relational structures reaches a preregistered structural-OOD capability target at lower total cost than* static Orion structural curation *and the strongest faithful model-aware/skill-aware parent, while satisfying registered composition, boundary, retention and negative-transfer constraints.*

The critical comparison is therefore

$$\boxed{\text{Adaptive Orion} - \text{Static Orion}},$$

not merely Orion versus random sampling. If this contrast is measured but indistinguishable, the learner-conditioned extension is not established even if static structural curation remains useful. If a simpler parent such as STAT- or MATES-style selection matches the result, the claim must narrow to any residual value contributed by the directional structural/boundary/shared-substrate machinery. No result for this extension is claimed in the present manuscript.

Each hypothesis has a direct falsifier. Flat structural coefficients, high Q3 transfer, no break-even cost, a strong MASS/Skill-It/MATES/STAT/SWIFT-like parent matching Orion, adaptive Orion failing to beat static Orion, harmful forgetting/composition trade-offs, or incompatible training/inference representations all narrow the paper.

7 Evaluation programme

7.1 Training arm

The first phase uses small open models or controlled transformers so that the structural-redundancy mechanism can be tested before expensive scaling. Training datasets are generated or curated at controlled structural-redundancy levels while token length, difficulty and surface-semantic diversity are held as stable as feasible. Baselines include uniform sampling, exact/semantic deduplication or diversity selection, Skill-It-style dependency sampling, MASS-style skill-graph selection, and at least one faithful modern model-aware adaptive parent where the task representation permits it. MATES is a primary parent for evolving model-aware data influence [3]; STAT is a primary parent for learner-specific missing-skill targeting and explicit saturation-aware adaptive selection/synthesis [6]. ADO and model-state-dependent spaced scheduling provide additional online/dynamic allocation controls [4, 5]. Rho-1 supplies a strong selective-loss/token-utility parent where its reference-model contract is applicable [7]. The Orion arm differs only when it uses cross-domain structural objects and witnessed structural coordinates that these parents do not already provide.

Primary outcomes are tokens/examples to a frozen structural-OOD capability target, total training compute where measurable, performance on structure-known/domain-new tasks, performance on genuinely structure-new tasks, calibration and negative transfer. A positive result on ordinary in-domain accuracy without structural OOD does not support the central claim. Skill-It can carry parent authority only if its data-driven dependency learning and sequential adaptive sampler are reproduced under matched model, corpus, token budget, evaluation schedule and seeds; static dependency overlap is not a faithful substitute. MASS is a mathematical-pretraining parent and requires a parent-compatible mathematical skill graph, selection rule and matched mathematical evaluation. MATES/STAT-style parents must likewise be reproduced with their model-state feedback or missing-skill mechanism intact; replacing them by a static score would artificially weaken the comparator.

7.2 Learner-conditioned extension: static versus adaptive structural allocation

The learner-conditioned extension is evaluated only after a cheap controlled exposure study shows that the proposed mastery coordinates are measurable. It explicitly separates existing Paper II static structural curation from a new adaptive treatment. Let the provisional learner state for registered structure S be

$$m_t(S) = (m_P, m_C, m_B, m_R, m_T, m_F),$$

for principle, composition, boundary, representation, transfer and retention coordinates. Each coordinate is computed from a frozen probe family and is bound to the exact model/checkpoint; no model self-report is sufficient.

The confirmatory matched arms are:

1. uniform/random;
2. strong semantic deduplication/diversity;
3. strongest faithful skill/model-aware parent for the task family (for example STAT, MATES, Skill-It or MASS under its native contract);
4. static Orion structural curation;
5. adaptive learner-conditioned Orion;
6. optionally, loss/perplexity-only and influence-only adaptive controls when resources permit.

The primary extension estimand is

$$\Delta_{\text{adaptive}} = \text{CTC}_{\text{static Orion}} - \text{CTC}_{\text{adaptive Orion}},$$

subject to registered non-inferiority/harm constraints for composition, boundary robustness, retention and invalid transfer. If Δ_{adaptive} is measured but indistinguishable, learner-conditioned structural saturation is not established. A positive adaptive-versus-random contrast cannot substitute for this test.

The first controlled experiment varies independently the underlying principle, surface form, numerical parameters, representation, application domain, boundary conditions and principle composition. Gold structural identity comes from the generator/verifier rather than an LLM judge. For one family the model is trained at increasing exposure counts; at each checkpoint the next equal-cost example is varied between another equivalent structural instance, a new representation, a new boundary, a new domain, a new composition and a hostile semantic near-miss. This distinguishes declining marginal value of equivalent exposure from still-unsaturated composition or boundary coordinates.

Repetition is not forced to zero after apparent mastery. D4 reports that intelligently repeated data can improve pretraining [8], and the training study therefore includes zero-revisit, low-floor, spaced/varied-revisit and continued-repetition controls. A strong adaptive result must save total cost without material forgetting, calibration loss or composition degradation. The scheduler is interpreted as reallocating *marginal* budget, not certifying that repeated exposure has no value.

The adaptive allocation policy is also subject to the same evidence-governed failure principle used elsewhere in Orion: a flat or harmful training result first produces competing typed diagnoses (for example same-structure saturation, composition gap, boundary gap, representation gap, transfer-domain gap, forgetting, model/optimization floor or instrument defect). A policy change is justified only after a bounded matched discriminator supports the diagnosis; the resulting scheduler is a challenger and must generalize on disjoint fresh structures before promotion. Failure therefore motivates a specific tested allocation change rather than another unconstrained curriculum idea.

7.3 Inference arm

The main inference experiment freezes the base model. Baselines include vanilla reasoning, semantic exemplar/memory retrieval, analogy prompting, a reusable reasoning-primitive library and, only where tasks expose a shared executable operator interface, a structural workflow prior in the style of SWIFT. The Orion arm retrieves a structural object, validates a directional witness, retrieves or adapts a validated operator where one exists, and verifies the target result. Outcomes include capability, Q2 transfer, Q3 invalid-transfer rate, reasoning tokens, latency, search nodes, tool calls and verification cost. A faithful SWIFT comparison matches the base model, operator library, source workflows, workflow-search budget, interface, verifier and execution budget and charges prior-construction cost. For scientific pairs without an executable workflow interface, SWIFT is marked not applicable rather than weakened into a graph-similarity scalar.

For the optional training-time extension, the strongest shared-substrate claim additionally requires exact registered structural identities to survive the train/inference boundary. The structure IDs, roles, relations, invariants and boundaries used to group or allocate training examples must be reusable by the inference-time transfer/witness layer. Recent work independently studies emergence and test-time use of learned structural information [12] and reusable latent experiences composed into adapters [13]; Orion therefore claims no novelty for train/test structural reuse in the abstract. If the training scheduler requires an unrelated latent skill representation, the result must be reported as two mechanisms rather than one shared Orion substrate.

7.4 Controlled structural redundancy

A synthetic or semi-synthetic generator produces datasets with increasing effective duplication of underlying structures while surface forms remain varied. The preregistered curve is

$$G(\rho_S) = \mathrm{CTC}_{\mathrm{parent}} - \mathrm{CTC}_{\mathrm{Orion}},$$

where ρ_S is an uncertainty-aware structural-redundancy measure and CTC is cost to the frozen capability target. The analysis tests a predeclared monotone or rank relationship rather than choosing a curve after observing results.

For the learner-conditioned extension, static ρ_S is insufficient. The same candidate may be useful early and redundant later, or may remain valuable because it introduces a new composition/boundary/representation despite sharing the base principle. The adaptive analysis therefore conditions allocation on the frozen learner/checkpoint state and reports the full coordinate-specific mastery/uncertainty record rather than one post-hoc scalar "mastery" threshold.

7.5 Parent-method assimilation

Each strong baseline is first treated as a parent method. Its strongest relevant mechanism is reproduced as faithfully as feasible and mapped into the Orion atlas. If the mechanism is useful, the final system may assimilate it; the scientific comparison then asks whether the residual Orion structure adds value beyond the parent. For example, MASS and Skill-It become skill/dependency training-selection parents; MATES becomes an evolving model-aware influence parent; STAT becomes a missing-skill/saturation parent; ADO/Sst become online allocation parents; SWIFT becomes a workflow-structure parent; and Reasoning Primitive Induction/TraceCompiler become procedural-reuse parents. The project does not intentionally weaken them to preserve novelty.

The cheap signal gate separately requires a modern semantic comparator. Its frozen control consumes the exact label-blind source and target descriptions with a hash-bound BGE cross-encoder while excluding proposed structural witnesses and all outcome fields. Missing or mismatched model files, content hashes, cases or pre-label chronology return `CANNOT_CHECK`; surface Jaccard cannot be substituted. No descriptor or result exists at this protocol-freeze stage because the exact model asset has not been staged.

Native execution is additionally frozen as two distinct LUNARC CPU batches. An allocated staging job must download the exact public revision and locally verify all six file sizes and SHA–256 values before atomic promotion. A separate descriptor batch is submittable only after the scheduler-bound stage harvest passes for the same exact clean merged checkout. It runs offline with local files only, requires the fast-tokenizer pair probe, and requires the shared runtime tree to equal the exact digest first attested by the allocated staging job and to remain unchanged through inference. A later descriptor harvest is authoritative only with a payload-free post-descriptor zero-label observation or a first-label cutoff proving that the descriptor timestamp precedes label availability. The freeze-time zero-label receipt is preflight evidence only. At this contract stage neither job has been submitted and no descriptor exists.

7.6 Stopping rule

Large-scale training is not authorized unless the low-cost mechanism tests pass. The programme stops or narrows if:

1. the structural score does not add transfer information beyond semantic similarity;
2. Q3 semantic decoys are frequently transferred;
3. structure labels are unstable across independent annotators or extraction methods;
4. a parent method matches the cost/capability frontier;
5. induction, mastery-probe, selection and verification cost eliminate any realistic break-even point;
6. adaptive Orion does not beat static Orion on the preregistered fresh structural-OOD estimand;
7. reduced repetition materially worsens retention, calibration, boundary robustness or composition generalization; or

8. the training and inference systems require unrelated representations.

A null pilot is therefore useful: it prevents a large compute spend on a mechanism that has not survived its own easiest falsifiers.

The label-visible v1 proposal is not reusable as the confirmatory benchmark even if it later receives external annotations. Training authorization is evaluated only from the separately frozen v2 protocol, fresh label-blind items, externally provenanced annotations and adjudication, and the v2 held-out signal receipt. The batch-submission preflight binds all of those hashes to the exact Git subject and fails before invoking `sbatch` when any gate is absent or false.

7.7 Orion fresh-transfer evaluation

The downstream v3 experiment separates memory persistence from transfer validity. Both arms use the same underlying model, tools and frozen development experiences. A semantic-control arm retrieves and reuses candidate past experience using the strongest feasible content model; the witness arm applies the directional structural/boundary gate before the same promoted lessons or operators can influence routing. Every fresh-transfer task starts independently from the same frozen post-development state so one transfer case cannot teach the next.

Primary outcomes are target-task success/score, invalid-transfer or false-lesson reuse, repeated-failure rate, Q2 true accept, Q3 false accept and total resource use. Hostile near-misses are mandatory because an always-reuse policy can improve repeated-family tasks while increasing out-of-scope failures. A positive result requires material value beyond the strong content control under the same target verification rules; target scientific correctness cannot be inferred from the witness decision itself.

This experiment is distinct from the optional training-data redundancy and learner-conditioned saturation hypotheses. Even if witness-gated experience reuse improves fresh inference, no claim about training-data efficiency follows without the separately authorized training experiment and full cost-to-capability accounting. Likewise, current external-state learning with frozen model weights does not establish any result for the optional weight-updating extension.

8 Engineering reporting contract

The empirical article must be interpretable as an AI-systems result, not only as a representation proposal. We therefore predeclare a reporting contract that couples capability, transfer validity and resource cost. The experimental unit is the task or evaluation item; repeated seeds are nested replicates. Principal comparisons freeze the base checkpoint, tokenizer, task set, held-out domains/families, context window, tool permissions, evaluator, budget schedule and seed schedule. Training-selection comparisons additionally freeze the downstream optimizer and training recipe; inference comparisons freeze the model weights.

8.1 Metrics that answer the engineering questions

For transfer discrimination, we report held-out ROC-AUC/PR-AUC, log loss or Brier score, calibration, Q2 true-accept and Q3 false-accept against the strongest semantic/skill control. The load-bearing quantity is incremental held-out value from witnessed structure, not raw accuracy of a structural classifier.

For training efficiency, let $T_m(q^*)$ be the minimum cumulative training-token exposure required by method m to reach a preregistered structural-OOD capability target q^*. The headline token reduction against parent p is

$$\Delta_T(q^*) = 1 - \frac{T_{\text{Orion}}(q^*)}{T_p(q^*)}.$$

Metric	Definition	What it tells an engineer
Invalid-transfer false-accept rate	verifier-invalid transfers the gate licensed, over all invalid transfers	whether the gate is truly fail-closed; the load-bearing safety number (target 0)
Abstention coverage	cases resolved to `CANNOT_CHECK`, over all cases	how often the gate honestly declines instead of guessing
Paired reduction vs. strongest control	held-out paired binary-Brier (or accuracy) reduction of the full contract over the best non-structural control, with bootstrap CI	whether witnessed structure adds value beyond the strongest baseline, not just beyond lexical similarity
Break-even count	smallest reuse count at which the structural layer is no costlier than the comparator; "no finite break-even" when reuse never pays	when reuse becomes economical for a workload
Tokens per valid solve (TPVS)	billable inference tokens over count of solves that also pass the transfer/verification gates (∞ at zero valid solves)	cost per *trustworthy* success, not per raw answer

Table 1: Transfer-governance metrics an implementer can instrument. Reported as a profile; each measures transfer discipline or cost rather than accuracy alone.

We report the same target-based comparison for examples and, where measurable, training compute, together with a normalized area under the learning curve over a frozen token range. Structural-OOD is split into tasks with known structure but a new domain and the harder boundary in which whole structure families are held out; ordinary in-domain performance is retained as a safety check.

For inference efficiency, a *valid solve* requires task correctness plus the registered transfer and verification gates. Billable inference tokens sum provider-reported input and output tokens across retrieval planning, adaptation, criticism and verification calls. For stratum s,

$$\mathrm{TPVS}(s) = \frac{\sum_{i \in s} \text{billable inference tokens}_i}{\sum_{i \in s} \mathbf{1}[\text{valid solve}_i]}.$$

Zero valid solves therefore produce an infinite rather than missing cost-per-success value. We additionally report provider cost, tool calls, retrieval operations, median and p95 latency, and verification overhead.

The amortization claim uses cumulative total cost rather than online tokens alone:

$$\begin{aligned} C_{\text{total}}(n) = {} & C_{\text{induction}} + C_{\text{training delta}} + C_{\text{storage}} \\ & + C_{\text{retrieval}}(n) + C_{\text{adaptation}}(n) + C_{\text{reasoning}}(n) \\ & + C_{\text{tools}}(n) + C_{\text{verification}}(n). \end{aligned}$$

The empirical break-even count is the smallest registered reuse count for which Orion is no more costly than the comparator while the capability target and invalid-transfer ceiling remain satisfied. A workload with no break-even in the registered horizon is reported as such.

Table 1 summarizes the portable quantities an implementer can instrument on their own cross-domain reuse system. They are reported as a profile, never a single score, and describe transfer discipline and cost, not raw accuracy.

8.2 Main display logic

The empirical manuscript is planned around six claims rather than six data tables. Figure 1 explains the shared structural object and matched training/inference design without making a performance claim. Figure 2 tests whether witnessed structure predicts valid transfer beyond semantic/skill controls, with Q2 acceptance and Q3 false acceptance shown together. Figure 3 reports structural-OOD performance versus cumulative training tokens and tokens/compute to a frozen capability target. Figure 4 reports the frozen-model valid-success versus inference-cost frontier, including tokens per valid solve and Q2/Q3 safety. Figure 5 reports cumulative total cost and the empirical break-even region after induction, retrieval, adaptation and verification are charged. Figure 6 reports directionality, QoI, boundary, evidence and verification ablations as effect sizes rather than a decorative feature inventory.

All quantitative displays use editable vector text, final-size uncertainty definitions and task-level source data. They contain no arrows pointing to data points, no opaque text boxes over data and no point-by-point prose annotations. Panel labels remain outside the data region. Method names are carried by fixed-axis labels or a shared legend rather than text placed over marks. Main figures show the shortest evidence chain needed for the central claim; per-model, per-family, cache/reuse, token-decomposition, calibration and sensitivity analyses are reserved for Extended Data unless they become load bearing after outcomes are observed.

8.3 Representation ablations and shared-substrate criterion

The structural object is only justified if its fields are load bearing. We therefore ablate directionality, QoI, target boundaries, evidence identity, explicit non-preserved properties and post-transfer verification, and compare against replacing the witness with semantic similarity or a domain-specific skill label. Primary ablation outcomes are Q2 acceptance, Q3 false acceptance, valid task success, tokens per valid solve and break-even count.

The strongest shared-substrate claim requires the same structural identity/object type to be used in both data selection and inference reuse. If training and inference require unrelated representations, the evidence supports two narrower engineering mechanisms rather than one unifying substrate. If the representation improves validity but never reaches a realistic total-cost break-even, the paper must describe it as a reliability mechanism rather than an efficiency mechanism.

The corresponding shared-substrate claim is falsified when an executed preregistered necessary condition fails. Missing confirmatory annotation is instead missing evidence: it leaves that claim unlicensed without converting an unrun test into a negative result.

The full machine-readable metric and figure contract is frozen before the model-level experiments are run.

Code, materials and AI-use disclosure

A public research artifact accompanies this paper. Frozen benchmark proposals, strong-control protocols, annotation schemas and machine-readable receipts are retained alongside the manuscript so that publication prose cannot silently replace missing evidence. Language models were used as research, coding and drafting tools; they are not authors. The constructed diagnostic labels are explicitly distinguished from genuinely external annotation, and no external judgement is represented as completed until a hash-bound response, adjudication and provenance audit exist. Author identity, affiliation and corresponding email are given on the title page. Funding, competing-interest, acknowledgement and licensing declarations remain author-supplied submission metadata and are not inferred by the repository.

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